

Código y versión / Code and version: 808152-01 Código identificador / Identifier code: 808152 Idiomas / Languages: INGLÉS Dimensiones / Size: 210 X 297 MM Tamaño mínimo fuente / Minimum font size: 8 PT Tamaño mínimo interlineado / Minimum lead: 8 PT	Colores CMYK / CMYK colours: Colores Pantone / Pantone colours: Leyenda (no reproducen) / Legend (Not to print): TROQUEL DIE-CUT COTAS MEASUREMENTS	Acabados / Finished: Fecha / Date: 29/01/26 12:30H AF: CF Versión AF / AF Version: 0 SI PUEDE LEER ESTE TEXTO, EL DOCUMENTO NO ESTÁ CORRECTAMENTE VISUALIZADO. IF YOU CAN READ THIS TEXT, THE DOCUMENT IS NOT CORRECTLY DISPLAYED. Los colores que aparecen en esta prueba podrían sufrir variaciones respecto al resultado final, dependiendo de la impresora o monitor donde se visualice. Los colores Pantone se deben verificar con carta física Pantone.
---	--	--

PACKAGE INSERT

1. NAME OF THE MEDICINAL PRODUCT

Hidroferol 0.266 mg soft capsules

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each capsule contains 0.266 mg of calcifediol monohydrate.

Excipients with known effect:

Each soft capsule contains 5 mg of ethanol, 22 mg of sorbitol (E-420) and 1 mg of sunset yellow (E-110).

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Soft capsule.

Orange soft gelatine capsule, oval, of 15 mm per 9 mm, containing a clear, low viscous and free from particles liquid.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Treatment of vitamin D deficiency (i.e., 25(OH)D levels < 25 nmol/L) in adults.

Prevention of vitamin D deficiency in adults with identified risks such as in patients with malabsorption syndrome, chronic kidney disease mineral and bone disorder (CKD-MBD) or other identified risks.

As adjuvant for the specific treatment of the osteoporosis in patients with vitamin D deficiency or at risk of vitamin D deficiency.

4.2 Posology and method of administration

Posology

Treatment of vitamin D deficiency and prevention of vitamin D deficiency in patients with identified risks: one capsule once a month.

As adjuvant for the specific treatment of osteoporosis: one capsule once a month. Higher doses may be necessary in some patients after analytical verification of the extent of the deficiency. In those cases, the maximum dose administered should not exceed one capsule per week. Once the plasma levels of 25(OH)D are stabilised within the desired range, treatment should be discontinued, or the frequency of administration lowered.

Hidroferol 0.266 mg soft capsules should not be administered with a daily frequency. The dose, frequency and duration of the treatment will be determined by the prescriber taking into account the plasma levels of 25(OH)D, type and condition of the patient and other comorbidities such as obesity, malabsorption syndrome, treatment with corticosteroids. Hidroferol 0.266 mg soft capsule is recommended when administration spaced in time is preferred.

Serum concentrations of 25(OH)D should be monitored after initiation of the treatment, usually after 3-4 months.

The potency of this medicinal product is sometimes expressed in international units. These units are not interchangeable with the units used to express the potency of cholecalciferol (Vitamin D) preparations (see section 4.4)

Patients with Renal Impairment

Use of Hidroferol in patients with chronic kidney disease should be accompanied by periodic monitoring of serum calcium and phosphorus, and hypercalcemia prevention (see section 4.4).

Elderly population

No overall differences in safety or efficacy were observed between geriatric patients and younger adults.

Paediatric population

The safety and efficacy of Hidroferol in children and adolescents below the age of 18 years have not yet been established. No data are available.

Method of administration

Oral administration

4.3 Contraindications

- Hypersensitivity to the active ingredient or to any of the excipients listed in section 6.1
- Hypercalcemia (serum calcium > 2.6 mmol/L), or hypercalciuria.
- Calcium lithiasis.
- Hypervitaminosis D.

4.4 Special warnings and precautions for use

Hypercalcemia and hyperphosphatemia

To obtain an adequate clinical response to oral administration of calcifediol monohydrate, an appropriate dietary calcium intake is also required. Therefore, to control the therapeutic effects, the following parameters should be monitored, in addition to 25(OH)D: serum calcium, phosphorus and alkaline phosphatase as well as urinary calcium and phosphorus in 24 hours. A decrease in serum levels of alkaline phosphatase normally precedes the onset of hypercalcemia. Once parameters are stabilized and the patient is under maintenance treatment, the abovementioned determinations should be performed regularly, especially for serum levels of 25(OH)D and calcium.

Renal impairment

To be administered with caution. Use of this drug in patients with chronic kidney disease should be accompanied by periodic monitoring of serum calcium and phosphorus, and hypercalcemia prevention. Transformation to calcitriol takes place in the kidney; thus, in case of severe renal impairment (creatinine clearance of less than 30 ml/min) a very significant reduction in the pharmacological effects may occur.

Heart failure

Special caution is required. The patient's serum calcium should be monitored constantly, especially in patients on digitalis, because hypercalcemia may occur and arrhythmias appear. Twice-a-week determinations are recommended at the beginning of treatment.

Hypoparathyroidism

1-alpha-hydroxylase is activated by parathyroid hormone. As a result, in case of parathyroid insufficiency the activity of calcifediol may decrease.

Kidney stones

Calcemia should be monitored, since vitamin D increases absorption of calcium and may aggravate the situation. In these patients supplements of vitamin D should be administered only if the benefits outweigh the risks.

Prolonged immobilization

In patients with prolonged immobilization it may be necessary to reduce the dose in order to avoid hypercalcemia.

Sarcoidosis, tuberculosis, or other granulomatous diseases

To be administered with caution, since these conditions lead to a greater sensitivity to the effect of vitamin D as well as to an increase of the risk of adverse effects at doses lower than the recommended dose. It is necessary to monitor serum and urinary calcium concentrations in these patients.

Interference with laboratory tests

Calcifediol may interfere with determination of cholesterol (Zlatkis-Zak method), leading to false increases in serum cholesterol levels.

Warnings on excipients

This medicine contains 5 mg of alcohol (ethanol) in each soft capsule. The amount in one capsule of this medicine is equivalent to less than 1 ml beer or 1 ml wine. The small amount of alcohol in this medicine will not have any noticeable effects. This medicine contains 22 mg sorbitol in each capsule. This medicine contains Sunset yellow (E-110) which may cause allergic reactions. International Units (IU) should not be used for determination of the dose of calcifediol as this could lead to overdosing. Instead, the dosing recommendation in section 4.2 should be followed.

4.5 Interaction with other medicinal products and other forms of interaction

- **Phenytoin, phenobarbital, primidone** and other enzyme inducers: enzyme inducers may reduce plasma concentrations of calcifediol and inhibit its effects by inducing its hepatic metabolism. For this reason, it is generally recommended to monitor plasma 25-OH-D levels when calcifediol is administered with antiepileptics that are CYP3A4 inducers in order to consider supplementation.
- **Cardiac glycosides:** Calcifediol can cause hyperglycemia, which can, in turn, enhance the inotropic effects of digoxin and its toxicity, producing cardiac arrhythmias.
- Drugs that decrease the absorption of calcifediol such as **cholestyramine, colestipol or orlistat**, which can result in decreased effects. It is recommended to space doses of these medicines and vitamin D supplements at least 2 hours.
- **Paraffin and mineral oil:** Due to liposolubility of calcifediol, the product can dissolve in paraffin and intestinal absorption may decrease. Using other types of laxatives or at least spacing doses is recommended.
- **Thiazide diuretics:** Co-administration of a thiazide diuretic (hydrochlorothiazide) with vitamin D supplements in patients with hypoparathyroidism may lead to hypercalcemia, which may be temporary or require the interruption of the treatment with the vitamin D analogue.
- Some antibiotics, such as **penicillin, neomycin** and **chloramphenicol** can increase calcium absorption.
- **Phosphate-binding agents such as magnesium salts:** Since vitamin D has an effect on phosphate transport in the intestine, kidney and bone, hypermagnesemia may occur. The dosage of agents that bind to phosphate shall be adjusted according to phosphate concentrations in serum.
- **Verapamil:** Some studies show potential inhibition of antianginal action, due to antagonism of their actions.
- **Vitamin D:** Co-administration of any vitamin D analogue should be avoided as additive effects and hypercalcemia can occur.
- **Calcium supplements:** Uncontrolled intake of additional preparations containing calcium should be avoided.
- **Corticosteroids:** They counteract the effects of vitamin D analogue drugs such as calcifediol.

Interaction with food and drinks

Food supplemented with vitamin D should be taken into account, since additive effects may occur.

4.6 Fertility, pregnancy and lactation

Pregnancy

No controlled studies with calcifediol in pregnant women have been performed. Studies performed in animals have shown toxicity for reproduction (see section 5.3). Do not use this medicine during pregnancy.

Breast-feeding

Calcifediol is excreted into breast milk.

The risk in newborns/ infants cannot be excluded. Maternal ingestion of high doses of calcifediol can produce high levels of calcitriol in milk and cause hypercalcemia in infants.

This medicine should not be used during breast-feeding.

Fertility

There are no data on the effect of calcifediol on fertility.

4.7 Effects on ability to drive and use machines

Calcifediol has no or negligible influence on the ability to drive and use machines.

4.8 Undesirable effects

Frequencies are assigned as follows: Very common (≥1/10); Common (≥1/100 to <1/10); Uncommon (≥1/1,000 to <1/100); Rare (≥1/10,000 to <1/1,000); Very rare (<1/10,000) and not known (cannot be estimated from the available data).

The adverse effects related to vitamin D are associated to increased levels of calcium when an excessive intake of Vit D may occur i.e. associated with overdose or prolonged treatment. The doses of vitamin D analogues required for hypervitaminosis vary considerably from one subject to another. The adverse reactions due to increased levels of calcium can occur initially or at a later stage (see section 4.9 Overdose).

Código y versión / Code and version:
808152-01

Código identificador / Identifier code:
808152

Idiomas / Languages:
INGLÉS

Dimensiones / Size:
210 X 297 MM

Tamaño mínimo fuente / Minimum font size:
8 PT

Tamaño mínimo interlineado / Minimum lead:
8 PT

Colores CMYK / CMYK colours:



Colores Pantone / Pantone colours:

Leyenda (no reproducen) / Legend (Not to print):



Acabados / Finished:

Fecha / Date: 29/01/26 12:30H

AF: CF Versión AF / AF Version: 0



Los colores que aparecen en esta prueba podrían sufrir variaciones respecto al resultado final, dependiendo de la impresora o monitor donde se visualice. Los colores Pantone se deben verificar con carta física Pantone.

The immune system

Unknown frequency (cannot be calculated from the available data): Hypersensitivity reactions (such as anaphylaxis, angioedema, dyspnoea, rash, localised oedema / local swelling, and erythema)

Metabolism and nutrition disorders:

Unknown frequency (cannot be calculated from the available data): Hypercalcaemia and hypercalciuria.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorization of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via Spanish Pharmacovigilance of Human Use Medicines reporting system.

4.9 Overdose

Symptoms:

Administration of vitamin D in high doses or for long periods of time may cause hypercalcemia, hypercalciuria, hyperphosphatemia and renal failure. As early symptoms of overdose, weakness, fatigue, drowsiness, headache, anorexia, dry mouth, metallic taste, nausea, vomiting, abdominal cramps, polyuria, polydipsia, nocturia, constipation or diarrhoea, dizziness, tinnitus, ataxia, rash, hypotonia (especially in children), muscle or bone pain and irritability may appear.

Among later symptoms of hypercalcemia the following are included: runny nose, itching, decreased libido, nephrocalcinosis, renal failure, osteoporosis in adults, growth retardation in children, weight loss, anemia, conjunctivitis with calcification, photophobia, pancreatitis, elevated blood urea nitrogen (BUN), albuminuria, hypercholesterolemia, increased transaminases (SGOT and SGPT), hyperthermia, generalized vascular calcification, convulsions, soft tissue calcification. Rarely, patients may develop hypertension or psychotic symptoms; serum alkaline phosphatase may decrease; electrolyte imbalances together with moderate acidosis can lead to cardiac arrhythmias.

In the most serious cases, where serum calcium exceeds 3 mmol/l, syncope, metabolic acidosis and coma may happen. Although symptoms of overdose are usually reversible an overdose might lead to kidney or heart failure. It is accepted that serum levels of 25-OH-cholecalciferol above 375 nmol/l may be associated with an increased incidence of adverse effects. Increased calcium, phosphate, albumin and urea nitrogen in blood as well as cholesterol and blood transaminases are typical of this kind of overdose.

Treatment:

Treatment of calcifediol overdose consists of:

1. Withdrawal of treatment (with calcifediol) and with any calcium supplement being administered.
2. Follow a diet low in calcium. Administration of large volumes of liquids, both orally and parenterally, is advisable to increase calcium excretion. If necessary, administer steroids and induced forced diuresis with loop diuretics such as furosemide.
3. If intake has occurred in the previous 2 hours, gastric emptying and forced emesis are advisable. If vitamin D has already passed through the stomach, a laxative (paraffin or mineral oil) can be administered. If vitamin D has already been absorbed, hemodialysis or peritoneal dialysis with a dialysis solution free of calcium can be performed.

Hypercalcemia derived from prolonged administration of calcifediol persists for approximately 4 weeks after discontinuation of treatment. Signs and symptoms of hypercalcemia are usually reversible. However, metastatic calcification can cause serious kidney or heart failure and death.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Vitamin D and analogues, ATC code: A11CC06

Mechanism of action

Vitamin D has two main forms: D2 (ergocalciferol) and D3 (cholecalciferol). Vitamin D3 is synthesized in the skin by exposure to sunlight (ultraviolet radiation) and is obtained from the diet. Vitamin D3 must undergo a two-step metabolic process to be active; the first step occurs in the microsomal fraction of the liver where Vitamin D is hydroxylated at position 25 (25-hydroxycholecalciferol or calcifediol); the second step takes place in the kidney where 1,25-dihydroxycholecalciferol or calcitriol is formed due to the activity of enzyme 25-hydroxycholecalciferol 1-hydroxylase; conversion to 1,25-dihydroxycholecalciferol is regulated by its own concentration, by parathyroid hormone (PTH) and by serum calcium and phosphate concentration. Other metabolites with unknown function exist. 1,25-dihydroxycholecalciferol is transported from the kidney to target tissues (intestine, bone and possibly kidney and parathyroid gland) by binding to specific plasma proteins.

Pharmacodynamic effects

Vitamin D increases absorption of calcium and phosphorus in the intestine and improves normal bone formation and mineralization and acts on three levels: Intestine: Vitamin D enhances absorption of calcium and phosphorus in the small intestine.

Bone: calcitriol enhances bone formation by increasing levels of calcium and phosphate and stimulates action of osteoblasts.

Kidney: calcitriol enhances tubular reabsorption of calcium.

Parathyroid glands: vitamin D inhibits the secretion of parathyroid hormone.

Clinical efficacy and safety

The efficacy and safety of calcifediol monohydrate 0.266 mg soft capsules was evaluated in a randomized, double-blind study in post-menopausal women with 25(OH)D levels <50 nmol/L). 303 subjects were randomized and 298 conformed the intention to treat population. Patients were treated with calcifediol monohydrate 0.266 mg/month (N=200) or cholecalciferol (N=98) at dose of 625 micrograms/month (25000 IU). In the calcifediol group 98 patients received treatment for 4 months, the remaining patients (N=102) and the cholecalciferol group received treatment for 12 months.

After 1 month, 13.5% of patients treated with calcifediol monohydrate achieved 25(OH)D levels greater than 30 ng/mL (75 nmol/L) and after 4 months, this percentage increased to 35%. Highest 25(OH)D levels with calcifediol monohydrate were achieved after 4 months of treatment, indicating a non-accumulative effect.

The table below shows 25(OH)D concentration increases from baseline in ng/mL, as mean values (SD).

	Calcifediol 266 µg	Cholecalciferol 625 µg
Baseline	12.8 (3.9)	13.2 (3.7)
Increases from baseline:		
Month 1	9.7 (6.7)	5.1 (3.5)
Month 4	14.9 (8.1)	9.9 (5.7)
Month 12	11.4 (7.4)	9.2 (6.1)

*Results are shown as mean (SD)

5.2 Pharmacokinetic properties

Absorption

Calcifediol is well absorbed in the intestine; approximately 75-80% is absorbed through this process.

Following oral administration of calcifediol, the maximum serum concentration of 25-OH-cholecalciferol is reached after 4 hours approximately.

Distribution

Calcifediol circulates in the blood bound to a specific α-globulin (DBP). Is stored in adipose tissue and muscle for prolonged periods. Storage in adipose tissue is less significant than vitamin D, due to its lower lipid solubility.

Metabolism and Biotransformation

Production of calcitriol from calcifediol is catalysed by the 1-alpha-hydroxylase enzyme, CYP27B1, located in the kidney and all vitamin D-responsive tissues. CYP24A1, located in these tissues, catabolises both calcifediol and calcitriol to inactive metabolites.

Elimination

Calcifediol half-life is around 18 to 21 days and it is primarily excreted in the bile.

5.3 Preclinical safety data

Effects in non-clinical studies were observed only at exposures considered sufficiently in excess of the maximum human exposure indicating little relevance to clinical use. High doses of vitamin D (4 to 15 times the recommended dose in humans) have proved to be teratogenic in animals, but there are few studies in humans. Vitamin D can cause hypercalcemia in pregnant women, which could lead to a syndrome of supravalvar aortic stenosis, retinopathy and intellectual disability in infants and newborn.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Ethanol, anhydrous
Medium chain triglycerides
Gelatin
Glycerol (E-422)
Sorbitol Solution (70%) (E-420)
Titanium dioxide (CI=77891, E-171)
Sunset yellow (E-110)

6.2 Incompatibilities

Not applicable

6.3 Shelf life

4 years

6.4 Special precautions for storage

This medicinal product does not require any special storage conditions.

6.5 Nature and contents of container

This medicine is packed in PVC/PVDC-Alu blisters containing 1, 3 or 5 capsules. Blisters are packed in a cardboard box.

6.6 Special precautions for disposal and other handling

No special requirements for disposal.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MANUFACTURER

HC Clover Productos y Servicios, S.L
C/A Alicante 8-10
Arganda del Rey
28500 Madrid
Spain
For
Faes Farma, S.A.
Spain.

8. PRODUCT REGISTRATION HOLDER

Eurodrug Laboratories (M) Sdn Bhd
T2-21-10, The Square, One City,
Jalan USJ 25/1C, 47650 Subang Jaya, Selangor

9. DATE OF REVISION

18 Aug 2025

53MY59E99.A00